

Google Home + Sinric Pro Integration

Voice-Based Smart Home Appliance Control using ESP8266/ESP32

1. Overview

This document describes the Google Home integration used in the IoT-Based Smart Home Automation project. Sinric Pro provides the cloud connection between the ESP8266/ESP32-based device and Google Home. A connected appliance such as a light or relay-controlled load can therefore be controlled using voice commands.

2. Integration Flow

Google Home → Sinric Pro → Internet → ESP8266/ESP32 → Relay Module → Appliance

3. Requirements

- ESP8266 or ESP32 development board
- Relay module
- Connected appliance/load
- Wi-Fi network
- Sinric Pro account
- Google Home app and compatible Google Home/Assistant device
- Arduino IDE with the required Sinric Pro library

4. Create a Device in Sinric Pro

1. Create or log in to a Sinric Pro account.
2. Create a smart-home device, such as a Switch, for the appliance.
3. Give the device a clear name, such as Living Room Light.
4. Save the device and obtain the Device ID, App Key, and App Secret required by the firmware.
5. Keep these credentials private; do not publish them in a public GitHub repository.

5. ESP8266/ESP32 Firmware

The Arduino firmware connects the controller to Wi-Fi and Sinric Pro. When Google Home sends an ON/OFF command through the linked Sinric Pro account, the device receives the command and changes the relay output accordingly.

Important: store App Key, App Secret, Wi-Fi password, and other private credentials securely. Use placeholders or a separate private configuration file before publishing code on GitHub.

6. Link Sinric Pro with Google Home

1. Open the Google Home app.
2. Select Add.
3. Select Set up device.

4. Select Works with Google Home.
5. Search for Sinric Pro.
6. Select Sinric Pro and link your Sinric Pro account.
7. Allow Google Home to discover the devices associated with the linked account.
8. Assign the discovered appliance to the desired Google Home room.

7. Voice Control

Action	Example command
Turn ON light	Hey Google, turn on the light.
Turn OFF light	Hey Google, turn off the light.
Turn ON fan	Hey Google, turn on the fan.
Turn OFF fan	Hey Google, turn off the fan.

8. Testing Procedure

1. Power the ESP8266/ESP32 and relay circuit safely.
2. Confirm that the controller connects to Wi-Fi.
3. Confirm that the device connects to Sinric Pro.
4. Open Google Home and verify that the appliance is available.
5. Give an ON or OFF voice command.
6. Verify that the relay changes state and the connected appliance responds.

9. Troubleshooting

- **Device not visible in Google Home:** verify that the Sinric Pro account is correctly linked and run device discovery again.
- **Voice command does not work:** use a clear device name and check that the device type is supported by Google Home.
- **Relay does not respond:** check the ESP GPIO assignment, relay logic level, wiring, and firmware callback.
- **Device is offline:** check Wi-Fi credentials, network availability, and the controller's serial-monitor output.
- **Security:** never expose App Key/App Secret or Wi-Fi credentials in public source code.

10. Project Repository

Recommended location: **Documentation/Google-Home-SinricPro-Integration.pdf**

11. Reference

The integration procedure follows the official Sinric Pro documentation for linking a Sinric Pro device with Google Home.

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